

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 19 February 2026

Team ID: LTVIP2026TMIDS72781

Project Name: Flight Booking App (MERN Stack)

Maximum Marks: 4 Marks

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web UI for users	React.js, HTML, CSS, JS
2	Application Logic	Backend business logic	Node.js, Express.js
3	Authentication	Secure login & token validation	JWT, bcrypt
4	Database	Store users, flights, bookings	MongoDB Atlas
5	External API	Payment Processing	Razorpay / Stripe
6	Email Service	Booking confirmations	Nodemailer
7	Infrastructure	Cloud Deployment	AWS / Render / Vercel

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	MERN Stack Framework	React, Node, Express, MongoDB
2	Security Implementations	Token-based authentication	JWT, bcrypt
3	Scalable Architecture	3-Tier Architecture	Cloud Deployment
4	Availability	High uptime hosting	Cloud Hosting
5	Performance	Optimized APIs & Indexed Queries	Axios, MongoDB Indexing